ALA-TOO INTERNATIONAL UNIVERSITY

Faculty of Engineering and Informatics

Calculus II Assignment (**25**%)

Power series representation of functions.

1. Show that $\dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n$ for $|x| < 1$.

2. Using 1, show $\dfrac{1}{1+x^2} = \displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $-1 < x < 1$.

3. Using the **derivative** of 1, show $\dfrac{1}{(1-x)^2} = \displaystyle\sum_{n=0}^{\infty} (n+1)x^n$ for $|x| < 1$.

4. Using 1 and **integration**, show $\ln(1+x) = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$.

5. Using 2 and **integration**, show $\tan^{-1} x = \displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$.

Last Date of Submission: Wednesday, April 23, 2025.

Do it in a group and submit a hard copy (only on A4 size paper).